FarmX

Empowering Indian Agriculture Through Technology

Hackathon Presentation 2026

By Jeevan M & Team

Problem Statement & Theme Alignment

20 MARKS

No Price Transparency
Farmers sell at 30-40% below market rates due to information asymmetry

Fragmented Ecosystem
No unified platform connecting farmers, dealers, transporters & laborers

Manual Farming
90% of small farmers lack access to IoT, weather data, or pest alerts

Financial Tracking Gap
Farmers have no tools for P&L tracking, leading to hidden losses

Theme Fit

SDG 1
No Poverty

SDG 2
Zero Hunger

SDG 8
Decent Work

SDG 9
Industry & Innovation

Innovation & Uniqueness

20 MARKS

IoT Smart Irrigation

Automated pump control based on real-time soil moisture sensors with manual/auto toggle

AI Price Prediction

ML-powered crop price forecasting with 'Best Sell Window' recommendations

Pest Alert System

Location-based pest threat detection with severity levels & treatment guides

4-Role Ecosystem

Unified platform for Farmers, Middlemen, Drivers & Workers — no other app does this

Technical Feasibility

15 MARKS

Tech Stack

Frontend	React + TypeScript + Vite
Styling	Tailwind CSS + shadcn/ui
Charts	Recharts (data viz)
PWA	Offline-first mobile app
State	React Context + Shared Store
Routing	React Router v6

Architecture

- Modular Component Design
  - 20+ reusable components
- Role-Based Dashboards
  - Separate views per user type
- Shared Data Store
  - Cross-role data sync
- PWA Ready
  - Works offline on any device

Impact on Farmers & Rural Livelihood

25 MARKS

40%
Better Prices
via transparent market data

30%
Water Saved
through IoT smart irrigation

4
Roles Connected
unified agri ecosystem

₹0
Cost to Use
free for all farmers

Farmer Income

Direct market access eliminates middleman exploitation. Real-time prices ensure fair deals.

Rural Employment

Worker hiring system creates transparent job market for agricultural labor.

Resource Efficiency

IoT irrigation & weather data reduce waste. Machinery rental cuts capital needs.

Accessibility, Language & Inclusivity

20 MARKS

PWA — Works on Any Phone

No app store needed. Install from browser. Works on low-end Android devices with 2G/3G networks.

Simple, Icon-Driven UI

Emoji & icon-based navigation. Minimal text dependency. Color-coded roles for instant recognition.

Multi-Language Ready

Architecture supports Hindi, Kannada, Tamil, Telugu. Regional language integration planned.

Offline-First Design

Core features work without internet. Data syncs when connectivity is available.

Voice Input Ready

Designed for future voice-based crop listing & search for non-literate users.

Inclusive for All Roles

Equal access for small farmers, women workers, local drivers. No digital literacy barrier.

Live Features Overview

Farmer	Sell crops, Smart irrigation, Weather, Pest alerts, Price prediction, P&L tracking, Hire workers, Rent machinery
Middleman	Search & buy crops, Track purchases, Hire drivers, Financial accounts, Machinery rental
Driver	View transport jobs, Accept deliveries, Track earnings, List vehicles for rent
Worker	Find farm jobs, Apply for work, Track wages, View job history

Thank You!

Let's revolutionize Indian agriculture together.

Idea: Jeevan M & Rihan | Development: Jeevan M

Supporting Indian Agriculture